

CASA Report

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1 Introduction

1.1 The Need for Explainability in Critical AI Systems

The integration of Artificial Intelligence (AI) systems across every economic and technological sectors has brought the emergence of studies focused on system safety, reliability, and accountability. As AI deployments increasingly intersect with real-world environments the demand for transparency becomes absolute. For instance, in the event of an autonomous vehicle collision, establishing liability requires answering a fundamental question: what specific visual features was the system attending to at the moment of the decision-making process?

To address these vulnerabilities, the field of Explainable AI (XAI) is dedicated to decoding the internal representations of these architectures, aiming to understand how models process, correlate, and interpret informations during inference. While MLLMs represent a significant leap in AI capabilities, reasoning over text alongside images, audio, and video; their architectural complexity introduces the "black-box" problem. In vision-language tasks, the internal mapping between visual inputs and textual outputs is deeply obfuscated. Consequently, traditional evaluation metrics fail to provide consistent and reliable insights into whether a prediction is visually grounded or merely the result of linguistic hallucinations.

To overcome this opacity, a new paradigm of visual interpretability is required. Token Activation Maps (TAMs) are emerging as a robust spatial grounding mechanism to determine exactly where the model focuses its attention. By extracting the activation energies associated with specific linguistic tokens and projecting them back onto the original pixel space, TAMs provide a quantifiable visual footprint. This allows researchers to track the model's gaze and verify that its textual responses are genuinely anchored to the correct visual information.

1.2 The "Clever Hans" Effect in MLLMs

As MLLMs achieve unprecedented scores on traditional Visual Question Answering (VQA) benchmarks, the need for these rigorous XAI methodologies becomes paramount. Standard benchmarks evaluate these systems almost exclusively based on their final textual output. However, recent literature raises severe concerns regarding the internal validity of these evaluations. As highlighted by Pacchiardi et al. [1], LLMs and multimodal architectures can frequently exploit superficial patterns, spurious cues, or biases within the prompt to generate correct answers, entirely bypassing the actual capability being tested.

This phenomenon, known as the *Clever Hans effect*, is particularly dangerous in computer vision: an MLLM might confidently affirm the presence of an object without actually "seeing" it in the pixel space, relying instead on linguistic priors or compliance bias. To mitigate this critical flaw, evaluating the textual output is no longer sufficient. It is strictly necessary to mathematically inspect the internal topological state of the model, forcing an alignment between the generated word and the localized physical pixels.

2 Theoretical Background

2.1 Architecture and Information Flow in Multimodal Large Language Models

The rapid evolution of Large Language Models (LLMs) has extended into the multimodal domain, giving rise to Multimodal Large Language Models (MLLMs). These architectures are designed to bridge the semantic gap between continuous visual signals and discrete linguistic tokens. As highlighted in recent comprehensive surveys on MLLM interpretability [2], these models rely on deep self-attention mechanisms to perform complex reasoning over heterogeneous inputs.

A typical MLLM architecture, such as the Qwen2-VL series utilized in this study, consists of three primary components: a Vision Encoder, a Modality Projection Module, and an LLM Backbone. The Vision Encoder processes the input image, dividing it into a grid of localized patches and projecting them into continuous visual embeddings. Subsequently, the Modality Projection Module acts as a cross-modal adapter, aligning the dimensional space of the visual embeddings with the text token embeddings. Finally, the LLM Backbone processes the combined multimodal sequence. The internal dynamics of this processing are highly complex; recent studies on information flow in Large Vision-Language Models (LVLMs) demonstrate how models transition from encoding redundant spatial features to isolating relevant task-specific semantics during reasoning tasks [3].

2.2 Token Activation Maps (TAMs) for Visual Interpretability

To resolve the opacity of multimodal representations and accurately map generative linguistic outputs back to the physical pixel space, we leverage the formal framework of Token Activation Maps (TAMs). While traditional Class Activation Map (CAM) methods are tailored for single-prediction networks, they show limitations when applied to autoregressive Multimodal Large Language Models (MLLMs) because of the progressive nature of token generation. During sequential inference, preceding context tokens generate redundant visual activations that cascade through the self-attention layers interfering with the visual attribution maps of subsequent target tokens.

To isolate the true, uncorrupted visual evidence of each generated word, the TAM framework decouples visual attribution from contextual noise through two core modules: Estimated Causal Inference (ECI) and the Rank Gaussian Filter (RGF).

2.2.1 Initial Activation Projection

Given an input image and a text query, the vision encoder and cross-modal projection layer yield aligned visual features $\mathbf{F}_v \in \mathbb{R}^{N \times D}$, where N is the number of spatial vision tokens and D is the embedding dimension.[1] During generation, the MLLM sequentially emits text tokens. Let $\mathbf{W}_k$ be the classifier weights associated with the k -th token in the vocabulary. The raw spatial activation map $\mathbf{A}_k \in \mathbb{R}^N$ for any given token (either a prompt token j or a generated answer token i) is computed by projecting the visual features onto the corresponding token classifier weights [1]:

$$\mathbf{A}_k = \max(0, \mathbf{F}_v \mathbf{W}_k^T) \quad (1)$$

This projection extracts the direct logit-level alignment of the visual tokens relative to the target vocabulary item, serving as the raw attribution baseline.[1]

2.2.2 Estimated Causal Inference (ECI)

Under the potential outcome model (POM) of causal inference, we model the observed activation map $\mathbf{A}_i$ of the i -th answer token as a combination of its pure causal effect (the original visual information) and the confounding interferences propagated from the preceding context.[3, 1] We define the accumulated contextual interference $E(\mathbf{A}_{c,i})$ as a weighted sum of the activation maps of all preceding context tokens [1]:

$$E(\mathbf{A}_{c,i}) = \sum_{k \in C_i} r_{k,i} \mathbf{A}_k \quad (2)$$

where C_i denotes the set of indices of all context tokens generated prior to step i (including prompt tokens and preceding answer tokens), and $r_{k,i}$ is the cross-modal textual relevance between the k -th context token and the current target token i , derived directly from the text-to-text token activations.[1] A small scaling offset $\epsilon = 10^{-6}$ is added to prevent numerical division issues, and the constraint $r_{k,i} = 0$ is enforced when $k = i$ to avoid self-canceling.[1]

The refined causal activation map $\mathbf{A}'_i$ is computed by subtracting the estimated interference, scaled by an optimal factor β_i [1]:

$$\mathbf{A}'_i = \max(0, \mathbf{A}_i - \beta_i E(\mathbf{A}_{c,i})) \quad (3)$$

The optimal scaling coefficient β_i is determined dynamically via ordinary least squares (OLS) minimization, capturing the exact linear relationship of the redundant context across the visual token grid [1]:

$$\beta_i = \arg \min_{\beta} \|\mathbf{A}_i - \beta E(\mathbf{A}_{c,i})\|_2^2 \quad (4)$$

By subtracting this optimized context footprint, ECI successfully mitigates spatial hallucinations induced by preceding semantic context.[1]

2.2.3 Rank Gaussian Filtering (RGF)

Despite causal debiasing, raw subtraction often introduces high-frequency salt-and-pepper noise and localized gradient artifacts.[1, 4] Traditional spatial filters (e.g., standard Gaussian or median filters) either excessively blur object boundaries or fail to eliminate structured noise in sharp transition zones.[4] To address this, TAM introduces the Rank Gaussian Filter (RGF), a non-linear local operator.[1]

For each pixel x in the binarized causal map $\mathbf{A}'_i$, a local sliding window N_x of size $n \times n$ is defined.[1] The filtered value $\text{RGF}(\mathbf{A}'_i(x))$ is computed by applying a custom 1D Gaussian kernel to the sorted ranks of the activations within N_x rather than their raw intensities [1]:

$$\text{RGF}(\mathbf{A}'_i(x)) = \sum_{y \in N_x} g(\text{rank}(\mathbf{A}'_i(y)), \text{rank}_{\text{med}}, \sigma_{\text{cv}}) \mathbf{A}'_i(y) \quad (5)$$

where $\text{rank}(\mathbf{A}'_i(y))$ represents the ordinal rank of the activation value at location y within the sorted array of the local neighborhood N_x , and $\text{rank}_{\text{med}} = (|N_x| + 1)/2$ is the median rank position.[1] The Rank Gaussian kernel g is formulated as [1]:

$$g(r, r_{\text{med}}, \sigma_{\text{cv}}) = \frac{1}{\sigma_{\text{cv}} \sqrt{2\pi}} \exp\left(-\frac{(r - r_{\text{med}})^2}{2\sigma_{\text{cv}}^2}\right) \quad (6)$$

The standard deviation of the kernel is dynamically controlled by the local coefficient of variation σ_{cv} (the ratio of the standard deviation of the activations in N_x to their local mean) [1]:

$$\sigma_{\text{cv}} = \frac{\sigma(\mathbf{A}'_i(y \in N_x))}{\mu(\mathbf{A}'_i(y \in N_x)) + \epsilon} \quad (7)$$

By modulating the window variance using σ_{cv} , the filter automatically dampens noise in uniform background areas (where σ_{cv} is low) while preserving crisp object boundaries in high-contrast zones (where σ_{cv} is high).[1] The resulting refined activation map $\mathbf{A}^*_i = \text{RGF}(\mathbf{A}'_i)$ provides a mathematically rigorous, clean, and causally grounded footprint of the model’s visual attention for each generated word.[1]

3 Methodology and Experimental Pipeline

To investigate the spatial grounding capabilities and hallucination tendencies of Multimodal Large Language Models (MLLMs), this study introduces this experimental framework. The methodology is designed to move beyond traditional text-only evaluation metrics by dynamically linking the model’s linguistic output to its internal spatial attention mechanisms.

3.1 The Test Dataset

The empirical foundation of this research relies on the **MS-COCO (Common Objects in Context)** dataset [4], specifically utilizing the **val2017** split. MS-COCO is renowned for its dense, pixel-level annotations and structural complexity, allowing to evaluate the framework using an high controlled set of images.

To make sure that the visual grounding is assessed on precise object contours rather than generalized bounding boxes, all selected instances must possess valid polygon segmentation masks. These binary masks serve as the absolute Ground Truth (GT) in our pipeline. However, rather than employing random sampling, a deterministic filtering algorithm was developed to curate a controlled subset of 100 images.

The images selection criteria follows the following principles:

- **Clutter Reduction:** Images containing more than four total ground-truth annotations are systematically discarded. This constraint guarantees that the baseline visual environment is not crowded, preventing the model’s attention maps from becoming artificially fragmented by dozens of minor, overlapping instances.
- **Object Prominence:** To ensure the target objects are visually distinct and geometrically suitable for topological evaluation, valid instances must occupy a substantial spatial footprint. Specifically, objects lacking a segmentation mask or possessing a bounding area of 5000 pixels or less are filtered out.
- **Target Isolation:** The final selected scenes must contain strictly one or two valid, prominent objects that pass the area filter. This guarantees the presence of a clear primary target (*Primary Object*) while occasionally retaining a secondary element, providing a controlled, predictable baseline for the visual processing engine.

By applying these criteria, the resulting dataset provides a controlled environment that allows the pipeline to force the Model to navigate a realistic scene and anchor its text generation to visual evidence, without the evaluation being skewed by microscopic or heavily occluded background objects.

Prompting Strategies

A core hypothesis of this research is that the linguistic structure of the query acts as a conditioning mechanism, fundamentally altering not only the generated text but the way the model develops its visual attention. To empirically measure this phenomenon, the experimental pipeline subjects the Multimodal Large Language Model (MLLM) to three distinct prompting methodologies, each imposing a different cognitive and structural load.

3.1.1 Binary VQA (Hard Prompt)

The Binary Visual Question Answering paradigm represents the standard, zero-shot interrogative approach commonly used in traditional benchmarks. The model is presented with a direct, closed-ended question regarding the presence of the target (e.g., *"Is there a [target] in this image?"*), forcing an impulsive "Yes" or "No" classification.

- **Advantages:** This method provides a fast, standardized baseline for visual recognition. It assesses the model’s spontaneous ability to detect an object without complex semantic conditioning, minimizing the impact of the user over the reasoning process.
- **Disadvantages:** Hard prompts are notoriously susceptible to *compliance bias* (or affirmative bias). MLLMs tend to blindly confirm the presence of queried objects, that might lead to an high hallucination ratio to comply to what the user is asking.

3.1.2 Spatial Chain of Thought (CoT)

Inspired by cognitive decomposition, this paradigm forces the model into a sequential reasoning path before emitting a final verdict. The prompt instructs the model to explicitly describe the scene and map out the spatial coordinates of major elements (e.g., *"Step 1: Look at the image and describe... Step 2: Check for [target]... Step 3: Answer..."*).

- **Advantages:** The sequential generation acts as an attention regularizer. By verbalizing the visual context before focusing on the specific query, the model should inherently filter out irrelevant background noise, heavily reducing the occurrence of scattered distractors.
- **Disadvantages:** The primary drawback is the risk of "verbosity-induced hallucinations". Extended reasoning paths can cause the language generation module to derail, losing its visual anchor and hallucinating spatial relations or objects simply to maintain textual fluency.

3.1.3 Multiple Choice Questions (MCQ)

The MCQ paradigm tests the discriminative power of the model under contextual semantic noise. The prompt asks the model to identify the prominent object among four options (e.g., *"A) [Distractor 1], B) [Target], C) [Distractor 2], D) [Distractor 3]"*). In positive scenarios, the distractors are dynamically and randomly sampled from COCO categories absent from the image according to the dataset annotations.

- **Advantages:** It provides a rigorous evaluation of comparative semantics, forcing the model to weigh visual evidence against multiple competing hypotheses rather than simply confirming a single suggestion.
- **Disadvantages:** The presence of explicit false options inside the prompt introduces a severe "Distraction Bias". The model might struggle to ignore the textual distractors, inherently leading to a higher rate of logical failures (Wrong Choice).

3.2 The 4-Gate Validation Framework for Spatial Grounding

To evaluate the current qualities of the models, a Validation Framework has been designed according to the following 4 steps:

Gate 1: Semantic Parsing and Token Anchoring

The first gate evaluates the linguistic correctness of the generated text. To analyze the model's textual output, the parser employs a rule-based heuristic approach rather than a computationally heavy Natural Language Processing (NLP) pipeline. **Lowercasing** is uniformly applied to both the generated output and the COCO target category to ensure a case-insensitive evaluation.

To handle the lexical variability inherently produced by the Chain of Thought (CoT) paradigm, the validation utilizes advanced **Regular Expressions (Regex)** rather than naive substring matching. The system determines textual success by scanning the text for the target noun while simultaneously enforcing a **proximity constraint**. Specifically, the regex engine ensures that explicit negative keywords (e.g., "no", "not", "without") do not appear within a 30-character window preceding the target word detection. This prevents the validation gate from being deceived by complex negative formulations or intermediate adjectives (e.g., "there is no indication of a dining table").

If the model hallucinates an absent object or incorrectly classifies a positive scenario, the inference is immediately terminated as a text-based failure (**FN_Text**). Upon textual success, the parser iterates through the tokenized sequence. To ensure methodological fairness across the evaluated paradigms and avoid granting the CoT reasoning loop an artificial statistical advantage, the spatial extraction process is strictly anchored to the *final* occurrence of the target token, representing the model's definitive conclusion rather than its intermediate reasoning steps.

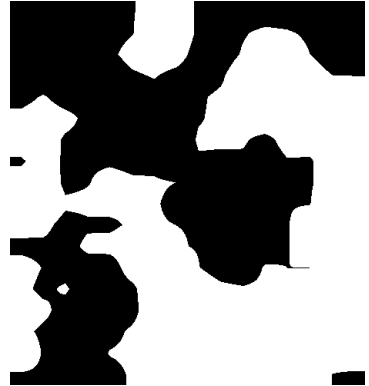
Gate 2: Energy Clustering via K-Means

Once the target token is anchored, its corresponding Token Activation Map (TAM) is extracted from the model’s intermediate hidden states. The raw heatmap represents the uncalibrated Softmax energy distributed across the image.

In early stages of this study, global binarization techniques such as Otsu’s thresholding method were tested to isolate the primary visual hotspot. However, Otsu’s method operates by maximizing inter-class variance under the assumption of a bimodal histogram. This approach proved inadequate for TAMs, as the Softmax function often generates a ”penumbra” or an attention halo, a smooth gradient of uncertainty spanning from the target to the background, resulting in multimodal energy distributions. Applying Otsu’s method produced maps characterized by unclear separation of focus levels or over-segmented patches, failing to cleanly isolate the true focal point, as showed in the following images.



(a) TAM HEATMAP



(b) OTSU RESULT IMAGE

Figure 1: In this example we can see how the Otsu method was not sufficient to separate light attention patches from the actual focus patches

To mathematically separate the model’s true focal core from the peripheral Softmax halo and the background, an unsupervised clustering technique was adopted evolving the Otsu method for a deeper separation. The normalized 2D heatmap pixels are reshaped into a one-dimensional feature space and processed using the K-Means algorithm ($K = 3$). These three distinct clusters are set to map to the semantic spatial regions: the background space, the penumbra, and the main focus of the model. The feature map is then binarized, isolating only the pixels belonging exclusively to the Core Focus cluster, ensuring a highly robust, topologically contiguous extraction of the model’s gaze.

Gate 3: Topological Distractor Filtering and Pointing Validation

The binarized Core Focus map undergoes morphological operations (closing) to bridge minor gaps, followed by contour extraction. Contours below a noise threshold are discarded. The remaining hotspots are evaluated topologically against the COCO Ground Truth (GT) segmentation mask:

- **Distractor Constraint:** Hotspots that do not intersect the GT mask are flagged as visual distractors. If the number of external distractors exceeds a tolerance threshold (set to 0 in this setup), the model’s attention is deemed too fragmented. The inference fails as *FP_Scattered*.
- **Centroid Pointing:** For the primary hotspot, the geometric centroid (c_x, c_y) is calculated using image moments ($M_{10}/M_{00}, M_{01}/M_{00}$). If the pointing centroid falls entirely outside the GT mask, it indicates that the model was looking at the wrong region despite mentioning the correct word, resulting in an *FP_Mispointed* failure.

Gate 4: Target-to-Background Ratio (TBR) Validation

The final gate evaluates the absolute visual confidence of the model by measuring the energy density contrast. The spatial bounding is quantified using the Target-to-Background Ratio (TBR), mathematically defined as:

$$TBR = \frac{\mu_{in}}{\mu_{out} + \epsilon}$$

where μ_{in} is the mean activation energy within the overlapping region of the Core Focus and the GT mask, and μ_{out} is the mean activation energy across the rest of the image background. A TBR value below a predefined acceptability threshold (τ_{TBR}), set a 1, indicates that the model’s visual confidence was too weak to be considered reliable because the model looked at the item as much as, or even less, than the rest of the image. Such instances are rejected as *FP_LowEnergy*. Only inferences that overcome this final energetic constraint are officially logged as successful visual groundings.

4 Experimental Setup and Implementation

The 4-Gate Validation Framework described in section 3.2 has been implemented using different software approaches. This section details the way the testing system was designed and implemented.

4.1 Model Architecture and Inference Parameters

The core evaluations were conducted using **Qwen2-VL-2B-Instruct** [5], an open-source Multimodal Large Language Model developed by Alibaba Cloud. This specific architecture was selected due to its native capability to process images of arbitrary resolutions and ratios dynamically, avoiding the aggressive cropping or resizing typical of older Vision Transformers (ViTs). Preserving the original spatial dimensions of the MS-COCO images is paramount for an accurate topological mapping of the Token Activation Maps (TAMs) against the ground-truth segmentation masks.

The model was instantiated using the HuggingFace **transformers** library in PyTorch. To maintain the highest fidelity in the activation energy, the model was loaded with standard 32-bit floating-point precision avoiding quantization techniques that could compress the hidden states and artificially alter the spatial distribution of the attention gradients.

During the inference phase, the generation hyper-parameters were controlled to balance determinism and reasoning capability across the three cognitive paradigms (Binary, CoT, MCQ). The specific configurations passed to the `model.generate()` function were:

- **Output Hidden States (`output_hidden_states=True`):** The most critical parameter for the Explainable AI (XAI) pipeline. This forces the model to expose its internal intermediate tensors, allowing the extraction of the multi-dimensional feature map ($f[-1]$) from the final Transformer layer immediately prior to the language modeling head projection.
- **Decoding Strategy (`do_sample=False`):** Greedy decoding was enabled in order to include only the options that the model considers as most probable, potentially reducing the number of hallucinations.
- **Token Constraints (`max_new_tokens`):** The generation limit was bounded to 15 tokens for Binary/MCQ prompts and expanded to 150 tokens for the CoT prompt to accommodate the step-by-step scene analysis.

4.2 Software Infrastructure and Automated Pipeline

To execute the 4-Gate Validation Framework across the curated MS-COCO dataset, a Python infrastructure was developed. Rather than relying on a single execution, the pipeline is divided into

three parallel evaluation scripts, each tailored to a specific prompt. Despite the differences in prompt handling, all scripts share a unified technological stack and a standardized validation pipeline.

4.2.1 Core Libraries and Dependencies

The implementation relies on an ecosystem of specialized libraries to handle deep learning inference, image processing, and data manipulation:

- **PyTorch and HuggingFace transformers:** Used as the primary deep learning backend to load Qwen2-VL-2B-Instruct, manage the tensor computations on the hardware, and extract the intermediate `hidden_states` generated during the forward pass.
- **TAM (Token Activation Maps):** The core extraction of the visual-linguistic grounding is powered by the TAM implementation provided by the xmed-lab (<https://github.com/xmed-lab/TAM>). This tool maps the 1D text tokens back to the 2D spatial features of the Vision Transformer.
- **OpenCV (cv2):** Essential for all topological operations. It executes the K-Means clustering algorithm, applies morphological kernels, and extracts geometrical contours via the Suzuki algorithm to isolate the focal hotspots.
- **Pycocotools & Pandas:** `pycocotools` is utilized to translate the polygon-based MS-COCO annotations into binary NumPy masks (the absolute Ground Truth). Pandas orchestrates the iterative loops over the dataset and tracks the final metrics for CSV exportation.

4.2.2 Pipeline Execution Flow and Code Sections

Each of the three experimental scripts follows a sequential pipeline. The workflow is structured into four distinct phases:

Phase 1: Scenario Initialization and Context Assembly At the start of each iteration, the script retrieves a target image. A randomized controller (Seed=12) determines the scenario type (75% Positive, 25% Negative). For Negative scenarios, the `pycocotools` API scans the JSON annotations to determine all categories present in the image and samples a target from a subset of categories from the COCO dataset that are absent in the annotations of the picked dataset. The corresponding prompt is then injected into the model’s chat template alongside the processed image tensor.

Phase 2: Inference and Semantic Token Anchoring The model performs the forward pass, returning the generated linguistic sequence. At this stage, the scripts diverge in their parsing logic to anchor the correct token:

- **Binary Script:** Scans the text for affirmative/negative structures and extracts the precise index of the primary noun.
- **CoT Script:** Scans the entire reasoning path and then collects the indices of *all* occurrences of the target word, evaluating the TAM for each mention to dynamically find the optimal spatial anchor with the highest confidence. This approach has been chosen to balance the more diffuse nature of CoT reasoning in which the model has been requested to explore the whole image before focusing on the target, augmenting the attention outside of the target mask. Choosing the map of the occurrence with the highest confidence allows the study to focus on the occurrence that represents the moment the model has gained the most knowledge about the presence or the absence of the target in the image.
- **MCQ Script:** It first isolates the chosen letter and then maps it back to the corresponding semantic token in the generated sequence.

Phase 3: Visual Extraction and K-Means Clustering Once the target token index is identified, the script interfaces with the TAM module to extract the corresponding attention map from the last hidden layer ($f[-1]$). The resulting heatmap is normalized between 0 and 1 using min-max scaling. To isolate the core focus, the 2D matrix is flattened into a 1D array of pixel values and fed into the `cv2.kmeans` function with $K = 3$ representing Background, Penumbra, and Core. The algorithm returns the cluster centers; the script identifies the cluster with the highest intensity center and generates a binary mask containing the pixels belonging to the main focus spots.

Phase 4: Morphological Validation and Metric Calculation The binary map undergoes a morphological closing operation using an elliptical kernel to bridge gaps. The script then computes the validation metrics necessary for the 4-Gate Framework:

- **Distractors :** Using `cv2.findContours`, individual hotspots are isolated. A *bitwise AND* operation checks the intersection between each hotspot and the `pycocotools` Ground Truth mask. Hotspots with a null intersection array are counted as external distractors.
- **Centroid Calculation:** For the primary hotspot, `cv2.moments` computes the spatial moments. The geometric centroid is calculated algorithmically as $c_x = M_{10}/M_{00}$ and $c_y = M_{01}/M_{00}$. The script verifies if the coordinate (c_x, c_y) falls inside the GT mask to avoid *FP_Mispointed* failures.
- **Target-to-Background Ratio (TBR):** Utilizing NumPy array masking, the script isolates the energy values. It computes the mean energy strictly inside the GT intersection (μ_{in}) and the mean energy on the background (μ_{out}). The TBR is calculated as

$$\frac{\mu_{in}}{\mu_{out} + 10^{-6}}$$

. The inference is appended to the Pandas DataFrame as a verified True Positive only if $TBR > 1.0$.

5 Results and Discussion

This section presents the results of the analysis of the data extracted by the 4-Gate architecture. The primary objective is to evaluate the three cognitive paradigms not only on the textual accuracy of the Qwen2-VL-2B-Instruct model, but more importantly, on its actual *visual grounding* capabilities.

Before proceeding with the numerical analysis, it is crucial to clarify the terminology adopted in this study. Within the context of our Validation Framework, traditional classification labels (TP, FP, TN, FN) are employed with an explicit abuse of notation. In this setup, a "Predicted Positive" (True Positive) requires both textual correctness *and* coherent spatial attribution. Consequently, instances where the text is correct but the Token Activation Map (TAM) is dispersed or poorly localized are categorized as "False Negatives" (representing *Scattered Attribution* or *Mislocalized Attribution* rather than pure classification errors).

5.1 Gate Survival Analysis

Analyzing the survival rate of the inferences across the validation pipeline allows us to observe the critical discrepancy between linguistic correctness and geometric validity. Figure 2 illustrates the percentage decay of inferences as they pass through the increasingly strict spatial criteria.

As shown in the graph, inferences retain near 100% survival through the *Textual Gate* (Phase 2), indicating that the model successfully generates the correct linguistic response across all three paradigms. However, a steep drop occurs at the *Geometric Gate* (Phase 3). This indicates that in the vast majority of cases, the model "guesses" the correct word without its attention being geometrically anchored to the correct Ground Truth pixels. Among the three methods, **Spatial CoT** and

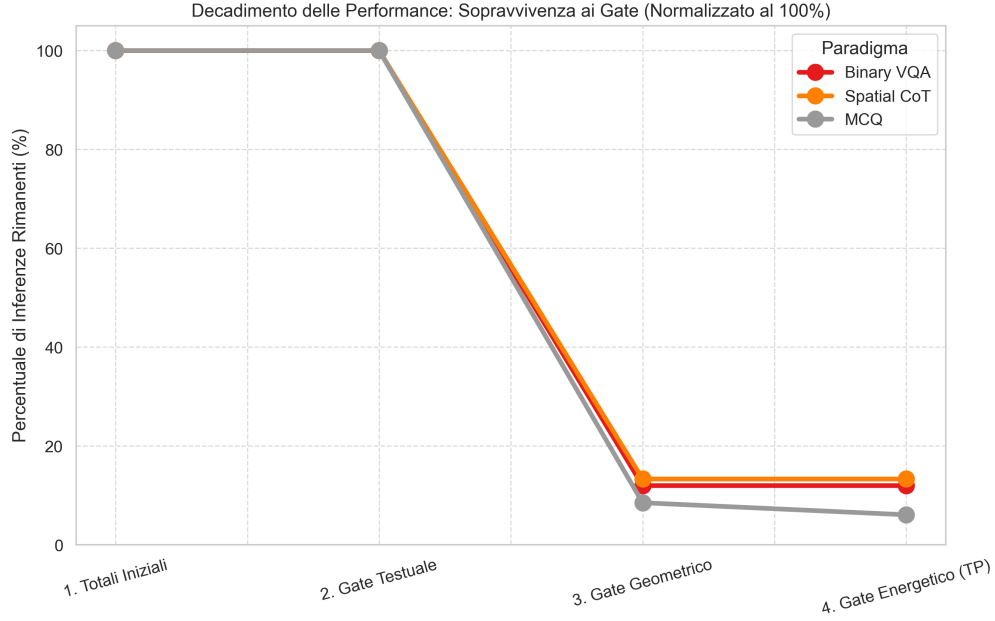


Figure 2: Performance decay (Gate Survival Rate) for positive queries normalized across the 4-Gate validation pipeline.

Binary VQA exhibit comparable resilience (surviving at $\sim 13.3\%$ and $\sim 12.0\%$ respectively). The multiple-choice (MCQ) paradigm suffers a near-total collapse, dropping to a 6.1% survival rate and demonstrating a severe inability to maintain visual grounding in the presence of textual distractors.

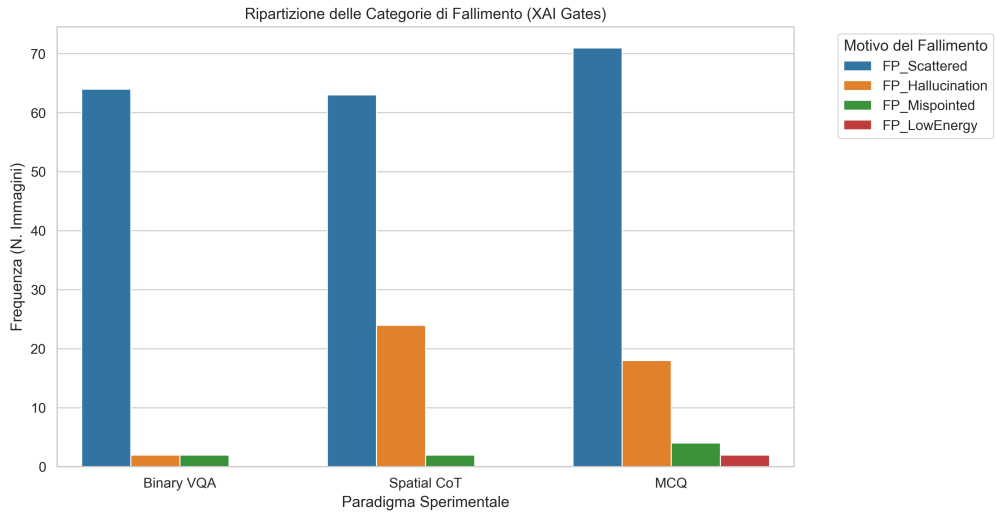


Figure 3: Distribution of specific failure categories across the XAI validation gates.

By analyzing the specific causes of failure (Figure 3), the dominant category across all configurations is **FP_Scattered**. This suggests that the primary limitation of MLLM attention is not total spatial misalignment (**FP_Mispointed**), but rather the severe diffusion of the visual focus. The Softmax function tends to distribute residual energy across multiple background elements. Notably, the Spatial

CoT and MCQ paradigms also register high frequencies of **FP_Hallucination**, confirming that complex or distractor-heavy prompts severely compromise the model’s visual purity.

5.2 Hallucination Mitigation and Classical Metrics

To evaluate the overall performance of the model, Confusion Matrices were calculated based on the visual-spatial validation criteria (Figure 4).

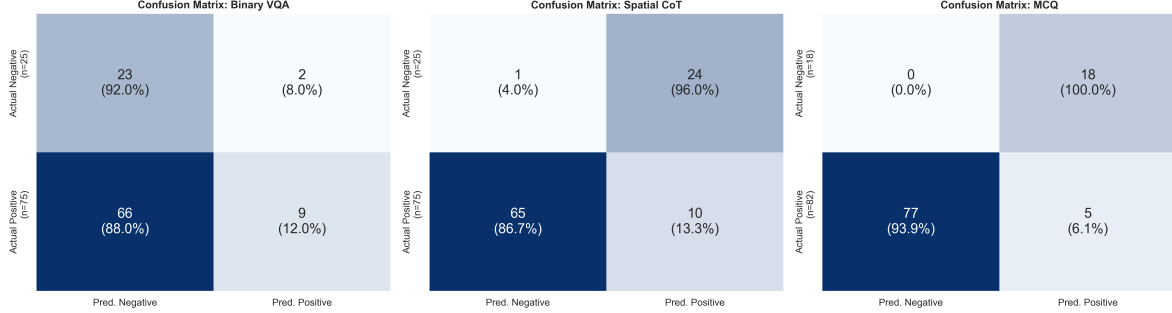


Figure 4: Confusion Matrices calculated using *Strict Visual Grounding* criteria.

Analyzing the matrices reveals divergent model behaviors and a significant trade-off between the paradigms. The **Binary VQA** setup demonstrates a conservative grounding approach: it achieves the highest number of True Negatives (23 out of 25 negative queries, 92.0%) and maintains an exceptionally low False Positive count (2 out of 25, 8.0%). This suggests that the baseline binary prompt exhibits a strong resistance to visual hallucination when the target object is absent according to COCO annotations.

Conversely, the **Spatial CoT** paradigm exhibits a different distribution, contradicting preliminary assumptions. While the step-by-step reasoning slightly increases the number of successfully grounded targets, this marginal gain in sensitivity is offset by a severe degradation in hallucination control. The Spatial CoT matrix indicates a sharp decrease in True Negatives (1 out of 25, 4.0%) and a massive spike in False Positives (24 out of 25, 96.0%). This pattern strongly indicates that forcing the model into an autoregressive spatial reasoning loop introduces a compliance bias, prompting the network to forcibly attribute visual energy to non-existent objects.

The **MCQ** paradigm demonstrates the weakest attribution quality overall. Yielding only 5 True Positives (6.1% out of 82 positive queries) and 0 True Negatives (0% out of 18 negative queries), alongside an overwhelming 77 False Negatives (93.9%), the data indicates that embedding textual distractors directly into the prompt severely disrupts the model’s interpretive coherence.

Translating these raw counts into classical performance metrics (Figure 5) highlights the underlying statistical dynamics. **Binary VQA** achieves vastly superior Accuracy and Precision, driven by its strict control over false positives. **Spatial CoT**, on the other hand, achieves a slightly higher Recall rate (13.3% vs 12.0%), indicating a marginally better ability to find the object when it is actually present. Unlike previous unseeded iterations, **Binary VQA** now clearly emerges as the most balanced approach, achieving the highest **F1-Score** (~ 0.21) driven heavily by its superior Precision. Spatial CoT falls behind in overall F1-Score (~ 0.18) due to its poor hallucination control, confirming that the step-by-step textual reasoning introduces severe noise into the spatial attribution.

5.3 Energetic Confidence Analysis

The final step involves measuring the intensity and purity of the model’s visual attention through the Target-to-Background Ratio (TBR), calculated exclusively on verified True Positives. The TBR effectively quantifies the visual “signal-to-noise ratio” of the architecture.

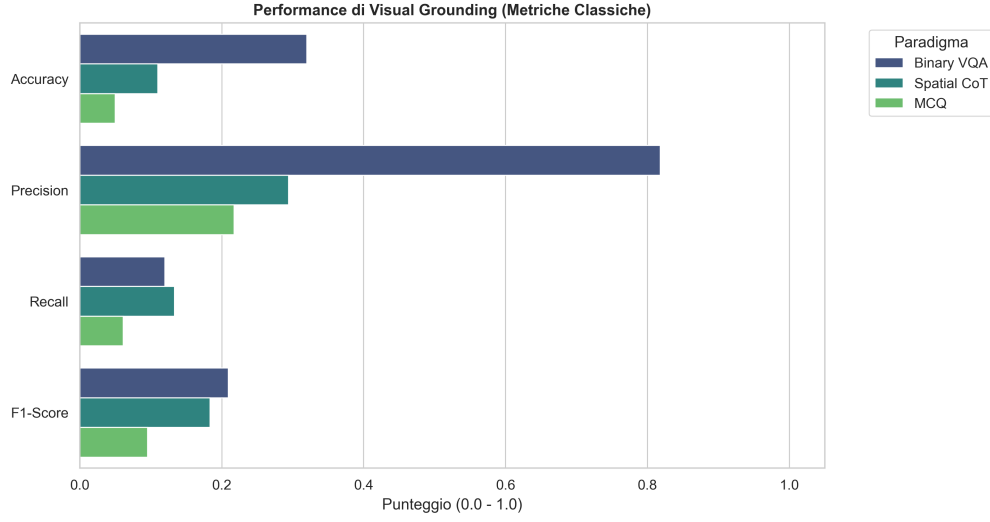


Figure 5: Comparison of classical evaluation metrics (Accuracy, Precision, Recall, F1-Score) strictly based on visual grounding.

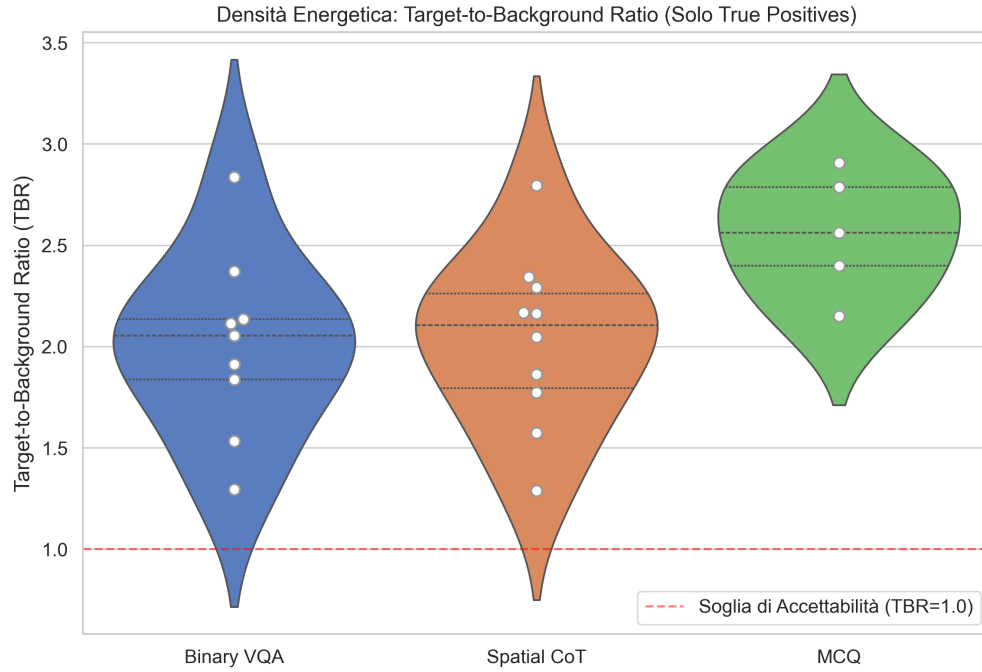


Figure 6: Density distribution of visual confidence (Target-to-Background Ratio) for verified True Positives.

The Violin Plot in Figure 6 provides evidence regarding the energetic distribution of the attention maps. It is important to note that these distributions are conditioned only on the few instances that survived the previous topological gates ($N = 9$ for Binary VQA, $N = 10$ for Spatial CoT, $N = 5$ for MCQ); therefore, broad generalizations must be made cautiously. Under the current deterministic setup, the **Spatial CoT** distribution suggests a slight upward shift in its median TBR

(~ 2.1) compared to Binary VQA (~ 2.0). This indicates that when the autoregressive reasoning path successfully grounds an object, the resulting activation energy tends to be slightly more isolated from the background noise, despite the paradigm’s overall tendency to hallucinate.

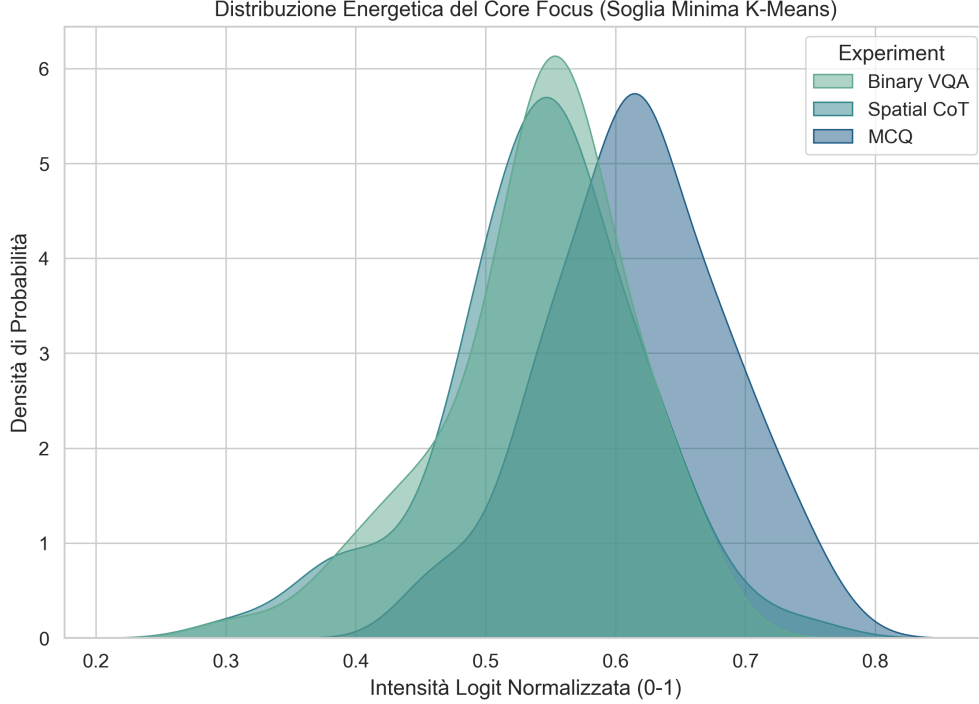


Figure 7: Probability density of the normalized Logit intensity within the Core Focus (K-Means Threshold).

This spatial intensity is further analyzed via the Kernel Density Estimation (KDE) of the attention score within the *Core Focus* (Figure 7). The plot illustrates that both Binary VQA and Spatial CoT achieve their highest probability density peaking around a threshold of 0.55, though Spatial CoT exhibits a slightly wider variance in attention intensity. The MCQ paradigm, inversely, shifts its distribution toward higher but highly unstable thresholds (~ 0.62), further reflecting the erratic spatial behavior induced by the *Distraction Bias* inherent to false options.

6 Conclusion and Future Work

6.1 Summary of Findings

This study set out to evaluate the spatial grounding capabilities of Multimodal Large Language Models (MLLMs), moving beyond text-only benchmarks. By analyzing the numerical results extracted through the 4-Gate Validation Framework, we can conclude that the current state-of-the-art, represented in this research by the Qwen2-VL architecture, is able to distinguish and describe elements within a complex image.

However, the empirical results demonstrate that this robust textual performance is frequently decoupled from visual comprehension. When inspecting the internal sighting process using Token Activation Maps (TAMs), the model’s visual focus often fails to align geometrically with the described objects. The high rate of attentional fragmentation indicates that MLLMs often rely on semantic compliance rather than visual physical evidence. At the same time the results may suggest that prompting

strategies, such as the Spatial Chain-of-Thought (CoT), can act as a regularizer, mitigating object hallucination and inducing confidence in the model in gaining visual proofs of the answers.

6.2 Rethinking Machine Attention: The Human-Centric Bias

A reflection brought up by these results concerns the nature of machine vision. The acceptability criteria developed in this study, where visual attention is considered valid only if bounded within the limits of the object, are built upon a human-based paradigm of visual cognition. Humans process scenes through foveal, highly localized focal points.

However, the way Transformer-based models "see" might be fundamentally different. The observed *FP_Scattered* failures and the energy distributed across the background may suggest that MLLMs might not require strict, localized attention to determine an object's presence. Instead, they may use a more diffuse, contextual attention mechanism, for example gathering statistical evidence from the surrounding environment rather than isolating the target item as we may believe. Understanding whether this scattered attention is a flaw or an alternative form of visual reasoning remains an open question that may lead to deeper research.

6.3 Future Directions

After this experimentation, I see different options for future research:

- **Architectural Scaling:** Investigating whether the spatial decoupling is a permanent structural flaw of the MLLMs or if it resolves by testing stronger, larger-parameter models.
- **Domain-Specific Fine-Tuning:** Applying the 4-Gate TAM pipeline to highly specialized models to evaluate if task-specific fine-tuning naturally enforces stricter visual grounding compared to general-purpose MLLMs.

6.4 Closing Remarks on Explainable AI (XAI)

Explainable AI is one of the most critical research topic of our time, in which Generative AI has taken the main stage across technological sectors. While the generative quality of these models is always increasing, their deployment in real-world environments, such as autonomous driving, robotics, and automated healthcare, cannot rely on textual accuracy alone. Ensuring that an AI system makes the right decision by looking at the right data is a mandatory step. Frameworks like the one proposed in this study represent the necessary bridge between the laboratory research and safe, reliable autonomous deployment.

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